Suggestion: Economics for Engineers
Unit 4: Depreciation, Replacement and Accounting

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1 Questions and Answers

1.1 What is depreciation? Explain the need for depreciation.

Ans: Depreciation refers to the gradual reduction in the value of an asset over time due to factors such as wear and tear, passage of time, technological obsolescence, and depletion. It represents the loss in usefulness and service potential of an asset during its working life.

The need for depreciation arises due to several reasons. First, it helps in ascertaining the true cost of production, as a portion of the asset value is charged against revenue each year. Second, depreciation ensures proper matching of cost with revenue, thereby determining accurate profit or loss. Third, it assists in capital recovery, so that sufficient funds are accumulated for replacement of the asset at the end of its life. Fourth, depreciation is necessary for correct valuation of assets in the balance sheet. Finally, depreciation reduces taxable income, leading to tax savings. Thus, depreciation plays a vital role in economic and financial decision making.

1.2 Explain the different types of property in the context of depreciation.

Ans: In the context of depreciation, property can be broadly classified into different types based on its nature and durability.

- Tangible property includes physical assets such as machinery, buildings, vehicles, and equipment. These assets are subject to depreciation because they wear out or become obsolete over time.
- Intangible property includes non-physical assets such as patents, copyrights, and goodwill. These assets lose value due to legal expiry or reduced usefulness and are amortized over their useful life.
- Wasting assets refer to natural resources such as mines, oil wells, and quarries. Their value diminishes due to extraction, and depreciation is calculated based on depletion.

This classification helps in selecting appropriate depreciation methods and ensures accurate economic evaluation of assets.

1.3 Explain the straight-line method of depreciation.

Ans: The straight-line method is the simplest and most widely used method of depreciation. Under this method, the asset loses an equal amount of value every year throughout its useful life.

Depreciation is calculated by subtracting the salvage value from the original cost of the asset and dividing the result by the estimated useful life. This results in a constant annual depreciation charge.

The straight-line method is easy to understand and apply. It is suitable for assets whose usage and benefits are uniform over time. However, it does not consider higher maintenance costs in later years and may not accurately reflect actual wear and tear.

1.4 Explain the diminishing balance method of depreciation.

Ans: The diminishing balance method, also known as the declining balance method, charges depreciation at a fixed percentage on the book value of the asset each year. As a result, depreciation is higher in the initial years and gradually decreases over time.

This method is suitable for assets that lose more value in the early years of their life, such as machinery and equipment subject to rapid technological change. It reflects the actual usage pattern more realistically than the straight-line method.

However, the method does not reduce the asset value to zero and salvage value is not directly considered. Despite this limitation, it is widely used for tax and financial reporting purposes.

1.5 Distinguish between straight-line method and diminishing balance method of depreciation.

Ans: The straight-line method charges equal depreciation every year, whereas the diminishing balance method charges higher depreciation in the initial years and lower depreciation in later years. Under the straight-line method, depreciation is calculated on the original cost, while under the diminishing balance method, depreciation is calculated on the book value. The straight-line method is simple and suitable for assets with uniform usage, whereas the diminishing balance method is suitable for assets that become obsolete quickly. Thus, the choice of method depends on the nature and usage of the asset.

1.6 Explain depreciation of a wasting asset.

Ans: A wasting asset is a natural resource whose value decreases as the resource is extracted. Examples include mines, oil wells, and quarries. Depreciation of such assets is known as depletion.

The cost of the asset is allocated based on the quantity of resource extracted during a period. As extraction progresses, the asset's value reduces until it is fully exhausted. This method ensures that depreciation is directly related to usage rather than time.

Depreciation of wasting assets is important for determining correct production cost and for planning replacement or closure of operations.

1.7 What is replacement analysis? Explain its importance.

Ans: Replacement analysis is the process of deciding whether an existing asset should be retained or replaced by a new one. It compares the economic performance of alternatives to determine the most cost-effective decision.

Replacement decisions arise due to increasing operating costs, declining efficiency, or technological obsolescence. Replacement analysis helps minimize total cost, improve efficiency, and ensure uninterrupted operation. It is an important tool for rational decision making in engineering economics.

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1.8 Explain the concepts of defender and challenger in replacement analysis.

Ans: In replacement analysis, the defender is the existing asset currently in use, while the challenger is the proposed new asset that may replace it.

The defender is evaluated based on its current market value, operating cost, and remaining life. The challenger is evaluated based on its initial cost, operating cost, and expected life. The decision to replace is made by comparing the economic performance of both alternatives.

This comparison ensures that replacement decisions are based on economic merit rather than historical cost.

1.9 What is meant by minimum cost life of an asset?

Ans: Minimum cost life of an asset refers to the period of service for which the average annual cost of owning and operating the asset is minimum.

As an asset ages, operating and maintenance costs increase, while capital recovery cost decreases. The minimum cost life occurs where the combined cost is lowest. This concept helps engineers determine the optimal replacement time for assets to achieve cost efficiency.

1.10 What is project appraisal? Explain its importance in engineering economics.

Ans: Project appraisal is the systematic evaluation of a project to determine its technical feasibility, economic viability, and financial soundness before investment decisions are made.

It helps in selecting the best project among alternatives, ensuring efficient allocation of resources, and minimizing risk. Project appraisal considers costs, benefits, risks, and long-term impact, making it an essential step in engineering economic decision making.

1.11 Distinguish between capital expenditure and revenue expenditure.

Ans: Capital expenditure refers to expenses incurred for acquiring or improving long-term assets, such as machinery or buildings, which provide benefits over several years. Revenue expenditure refers to expenses incurred for day-to-day operations, such as maintenance and repairs.

Capital expenditure increases the earning capacity of the business, whereas revenue expenditure is charged against current income. Proper classification is important for accurate financial and economic analysis.

1.12 Explain capital expenditure with suitable examples.

Ans: Capital expenditure is expenditure incurred to acquire fixed assets or to enhance their productive capacity. Examples include purchase of machinery, construction of buildings, and installation of new equipment.

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Such expenditure provides long-term benefits and is capitalized in accounting records. In engineering economics, capital expenditure decisions are evaluated carefully due to their long-term financial impact.

1.13 Explain revenue expenditure and its significance.

Ans: Revenue expenditure refers to expenses incurred for the maintenance and smooth operation of assets, such as repairs, wages, and utility costs. These expenses do not increase asset life or capacity.

Revenue expenditure is charged entirely to the current period and affects profitability. Proper control of revenue expenditure is essential for efficient cost management and economic operation.

1.14 Define depreciation. Explain the causes of depreciation with reference to deterioration and obsolescence.

Ans: Depreciation is the gradual and permanent decrease in the value of an asset over time due to use, passage of time, or other economic factors. It represents the loss in service potential of an asset during its useful life.

The main causes of depreciation are:

- Deterioration: This occurs due to physical wear and tear, corrosion, fatigue, and aging of materials as the asset is used. Continuous operation and environmental exposure reduce efficiency and value.

- Obsolescence: This occurs when an asset becomes outdated due to technological advancement, change in demand, or improved methods of production, even if the asset is physically sound.

Both deterioration and obsolescence reduce the economic usefulness of assets and must be considered for accurate cost and investment analysis.

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1.15 Distinguish between deterioration and obsolescence.

Ans: Deterioration refers to the physical decline of an asset due to wear and tear and aging, whereas obsolescence refers to loss of usefulness due to external economic or technological changes. Deterioration depends on usage and time, while obsolescence depends on innovation and market conditions. An asset may be obsolete even if it is in good physical condition. Both affect depreciation but arise from different causes.

1.16 Why is depreciation considered an expense?

Ans: Depreciation is considered an expense because it represents the portion of asset cost consumed during an accounting period. Assets are used to generate revenue, and depreciation allocates the cost of these assets over their useful life.

By charging depreciation as an expense, the matching principle is satisfied, ensuring that costs are matched with the revenues they help generate. This leads to accurate measurement of profit and prevents overstatement of income.

1.17 Why is depreciation regarded as a non-cash cost?

Ans: Depreciation is a non-cash cost because no actual cash outflow occurs when depreciation is charged. The cash payment occurs at the time of asset purchase, not during depreciation allocation.

Although depreciation reduces accounting profit, it does not reduce cash balance. However, it provides tax benefits and helps in capital recovery planning, making it economically significant despite being non-cash.

1.18 Explain capital allowance and its relationship with depreciation.

Ans: Capital allowance refers to the amount permitted by tax authorities as a deduction on capital assets for tax calculation purposes. It is closely related to depreciation but may differ in rate and method.

While depreciation is calculated for accounting and economic analysis, capital allowance is governed by tax laws. Capital allowance reduces taxable income and encourages investment in capital assets.

1.19 Explain the advantages of the straight-line method of depreciation.

Ans: The straight-line method has several advantages:

- Simple and easy to apply
- Equal depreciation charge every year
- Suitable for assets with uniform usage
- Helps in stable profit reporting
- Easy comparison of performance over years

Due to its simplicity and clarity, it is widely used in engineering economic analysis.

1.20 Explain the advantages of the diminishing balance method of depreciation.

Ans: The diminishing balance method offers the following advantages:

- Higher depreciation in early years
- Matches higher repair and maintenance costs in later years
- Suitable for assets subject to rapid technological change
- Better reflection of actual asset usage

It is particularly useful where assets lose value faster in the initial years.

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1.21 "Almost all tangible property can be depreciated." Explain.

Ans: Tangible property includes physical assets such as machinery, buildings, and equipment. These assets experience wear, aging, or obsolescence over time, resulting in loss of value.

Since they provide service over a limited life, their cost must be allocated systematically through depreciation. Hence, almost all tangible assets are depreciable except those that do not lose value, such as land.

1.22 "Land is never depreciated." Explain this statement as an exception.

Ans: Land is considered an exception to depreciation because it does not lose value due to use or passage of time. In many cases, land appreciates rather than depreciates.

Unlike machinery or buildings, land has an unlimited life and does not become obsolete. Therefore, depreciation is not charged on land in economic or accounting analysis.

1.23 Define book value of an asset. Explain its significance.

Ans: Book value is the value of an asset as shown in accounting records, calculated as original cost minus accumulated depreciation.

It is significant because it:

- Shows remaining asset value
- Helps in replacement decisions
- Aids in financial reporting
- Reflects asset consumption over time

Book value is widely used in economic and replacement analysis.

1.24 Define salvage value and scrap value. Distinguish between them.

Ans: Salvage value is the estimated value of an asset at the end of its useful life, whereas scrap value is the value obtained from selling the asset as raw material.

Salvage value is generally higher and planned, while scrap value is residual and minimal. Both are important in depreciation and replacement decisions.

1.25 What is meant by useful life of an asset?

Ans: Useful life is the period during which an asset can be economically used to provide service. It may end due to wear, obsolescence, or inefficiency.

Useful life may differ from physical life and plays a key role in depreciation calculation and asset management decisions.

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1.26 Explain the concept of economic life of an asset.

Ans: Economic life is the period during which the asset is cost-effective to operate. Beyond this period, operating and maintenance costs outweigh benefits.

Economic life may be shorter than physical life and is critical for replacement and investment analysis.

1.27 Explain the decision criteria for replacement of an existing asset.

Ans: An asset should be replaced when:

- Operating and maintenance costs rise significantly
- Efficiency decreases
- Better alternatives are available
- Total cost exceeds minimum economic cost

Replacement decisions aim to minimize long-term cost and improve efficiency.

1.28 Distinguish between marginal cost and average cost in replacement analysis.

Ans: Marginal cost refers to the additional cost of keeping an asset for one more period, whereas average cost is the total cost divided by service life.

Replacement is recommended when marginal cost exceeds average cost, indicating inefficiency.

1.29 Explain the replacement analysis decision map.

Ans: The replacement decision map provides a systematic procedure for comparing existing and new assets. It evaluates costs, service life, and efficiency to decide whether to retain or replace an asset.

This structured approach ensures rational and economical decisions.

1.30 Explain the different methods of calculating depreciation.

Ans: Common methods include:

- Straight-line method
- Diminishing balance method
- Sinking fund method
- Units of production method

Each method suits different asset usage patterns and economic objectives.

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1.31 Write a short note on expenditure on non-depreciable assets.

Ans: Non-depreciable assets, such as land, do not lose value over time. Expenditure on such assets is capitalized but not depreciated.

These assets are shown at historical cost and contribute to long-term stability rather than expense allocation.

1.32 What is a patent? Explain its treatment in depreciation context.

Ans: A patent is an intangible asset granting exclusive rights for an invention for a fixed period. Since patents have limited legal life, their cost is amortized over their useful period.

This ensures proper allocation of cost and accurate profit measurement.

1.33 Explain the role of GST relief in reducing overall tax burden.

Ans: GST relief reduces tax burden by allowing input tax credit, avoiding multiple taxation, and simplifying tax structure.

This lowers production cost, improves cash flow, and enhances economic efficiency in engineering projects.

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